Research Article

Indications, Trends, and Outcomes of Nerve Blocks for Spasticity Management: A Systematic Review

NeuroRehabilitation
2026, Vol. 58(2) 175–187
© The Author(s) 2026
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/10538135251413672
journals.sagepub.com/home/nre
Mary Ann Liebert
A Part of Sage

Sagarika Muradia[1] and Rama Prasad[1]

Abstract

Background: Spasticity is a stretch reflex disorder, clinically characterized by increase in muscle tone and is often seen in upper motor neuron lesions, which can limit function in daily activities. While botulinum toxin injections are widely used treatment option, nerve blocks may offer complementary or alternative benefits.

Objective: To evaluate the indications, techniques, and clinical outcomes of nerve blocks for spasticity management.

Methods: A literature search of major databases from July 2000– July 2025, identified studies on diagnostic nerve blocks, therapeutic neurolysis, and cryoneurolysis. Fourteen studies met inclusion criteria. Extracted data included patient characteristics, targeted nerves, agents and volumes, imaging guidance, tone reduction, functional outcomes, predictive value for BoNT, and adverse events. Due to heterogeneity in outcomes and methodologies, results were narratively synthesized.

Results: Diagnostic nerve blocks consistently aided treatment planning; one case–control study showed higher goal attainment (≈70% vs 40%) when diagnostic blocks guided BoNT injections. Phenol or alcohol neurolysis and cryoneurolysis produced meaningful spasticity reduction, typically improving Modified Ashworth Scale scores by 1–2 grades with effects lasting 3–6 months, often longer than BoNT. Across modalities, adverse events were uncommon and generally mild, most often transient pain or dysaesthesia. Publication trends showed increased use of ultrasound guidance and greater interest in diagnostic blocks and cryoneurolysis after 2018.

Conclusion: Nerve Blocks are effective yet underused techniques that can provide rapid and sometimes durable spasticity reduction and support individualized long term rehabilitationplanning. Further standardized, long-term studies are needed to optimize their integration into multimodal spasticity care.

Keywords

spasticity, brain injury , cerebrovascular disease, functional assessment, traumatic brain injury

Received: October 16, 2025; accepted: December 20, 2025

Introduction

Spasticity is a stretch reflex disorder, clinically characterized by increase in muscle tone that becomes more pronounced with rapid stretching movements (Corazza, 2020). It is a common consequence of lesions that damage upper motor neurons, causing upper motor neuron syndrome (UMNS) (Trompetto et al., 2014). Effective management of spasticity is therefore crucial for neurorehabilitation and to improve long-term functional outcomes.

A wide range of therapeutic options has been studied including oral medications, intrathecal therapies, botulinum neurotoxin (BoNT) injections, physical therapies, and surgical interventions (Banawal et al., 2025; Royal College of Physicians, 2018). Although BoNT is considered the standard treatment for focal spasticity, it is shown to have several limitations. Response can be variable and is shown to last for 3–4 months. (Doan et al., 2021; Royal College of Physicians, 2018). Dose ceilings and high cumulative costs can limit the feasibility of BoNT-A, with meta-analytic evidence demonstrating restricted effective dosing ranges (Doan et al., 2021) and NICE analyses showing substantial annual resource burden from repeated 12-weekly injection cycles (NICE, 2023). In rare cases of

[1]Neuro Rehabilitation Department, University Hospitals of Leicester NHS Trust, Leicester, UK

Corresponding Author:
Sagarika Muradia, Neuro Rehabilitation Department, University Hospitals of Leicester NHS Trust, Gwendolen Road, Leicester LE5 4PW, UK.
Email: S.muradia@nhs.net

antibody-related treatment failure have shown further constraints in its utility (Heinen et al., 2010).

Nerve blocks can be considered as a viable option of spasticity management because they offer both therapeutic and diagnostic benefits, provide immediate focal tone reduction and may be particularly useful when botulinum toxin might not be the best option due to the above reasons (Kocabas et al., 2010). Neurolytic agents such as phenol or alcohol act through axonal degeneration and demyelination (Kocabas et al., 2010), while cryoneurolysis, a newer method of performing nerve blocks, induces a reversible conduction block via axonotmesis (Winston et al., 2023). More recently, the use of diagnostic blocks with local anaesthetics, combined with ultrasound and/or electrical stimulation guidance, has shown their beneficial role in guiding treatment planning, improving injection targeting, and also predicting response to BoNT (Lam et al., 2015; Picelli et al., 2020).

Despite decades of clinical use, nerve blocks remain underutilized in spasticity management compared with other pharmacological and neurosurgical interventions. Variable techniques, non-standardised outcome measures, and lack of standardized protocols have limited their widespread adoption (Akkaya et al., 2010; Viel et al., 2002). Given the recent advances in use of technology and introduction of newer modalities such as cryoneurolysis along with exploration of different indications, a systematic synthesis of contemporary studies is timely to clarify their clinical role and inform future practice (Filippetti et al., 2024a; Lam et al., 2015; Lee & Jang, 2012; Winston et al., 2023).

The systematic review aims to evaluate the indications, techniques, and outcomes of nerve blocks used in the management of spasticity. Our goal was to assess trends in practice, commonly targeted nerves, specific agents used and clinical outcomes such as tone reduction, functional improvement, and predictive value for BoNT response and thus clarify the current role of nerve blocks in spasticity management and highlight areas where further research is needed.

Methods

Protocol and Registration

This systematic review was conducted in line with the 2020 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The review article was developed and registered with PROSPERO [CRD420251130441].

Eligibility Criteria

Study Design	Human studies published from 2000 onward; RCTs, cohort studies, case-control studies, case series (≥10 participants)
Population	Adults (≥18 years) with spasticity secondary to CNS lesions (e.g., acquired brain injury, spinal cord injury, multiple sclerosis)
Interventions	Nerve blocks for therapeutic or diagnostic use: phenol or alcohol neurolysis, diagnostic/local anaesthetic blocks, cryoneurolysis
Outcomes Reported	Spasticity reduction (e.g., Modified Ashworth Scale(MAS)), functional improvements in Activities of Daily Living(ADL),Goal attainment scale(GAS), predictive value for botulinum toxin dosage and use, safety/adverse events
Exclusion Criteria	Studies on children/adolescents (<18 years), surgical neurotomy, rhizotomy, intrathecal therapies only; case series <10 patients; single-patient case reports; narrative reviews; conference abstracts without full text.

Information Sources and Search Strategy

Electronic literature search using PubMed/MEDLINE, Embase (Ovid), CINAHL (EBSCO), and the Cochrane Library, covering studies published from January 2000 to July 2025 was conducted. Our search combined both controlled vocabulary (such as MeSH and Emtree terms) and free-text keywords related to spasticity, nerve blocks, neurolysis, phenol, alcohol, diagnostic nerve blocks, and cryoneurolysis. The full search strategies for each database are available in Supplementary Appendix 1.

Study Selection

The lead author (S.M) independently screened the titles and abstracts, the two researchers carefully reviewed the full texts of studies that seemed relevant independently. Discrepancies were discussed and resolved through consensus. The study selection process is shown in the PRISMA flow diagram (Figure 1).

Data Extraction

Data extraction was conducted using a standardized form that recorded key variables, including study characteristics, intervention details(type of block, nerve targeted, agent used, guidance method (landmark, ultrasound, stimulation)) and outcomes(spasticity measures (e.g., MAS), functional outcomes, predictive outcomes, and adverse events). Data extraction was performed independently by the two authors.

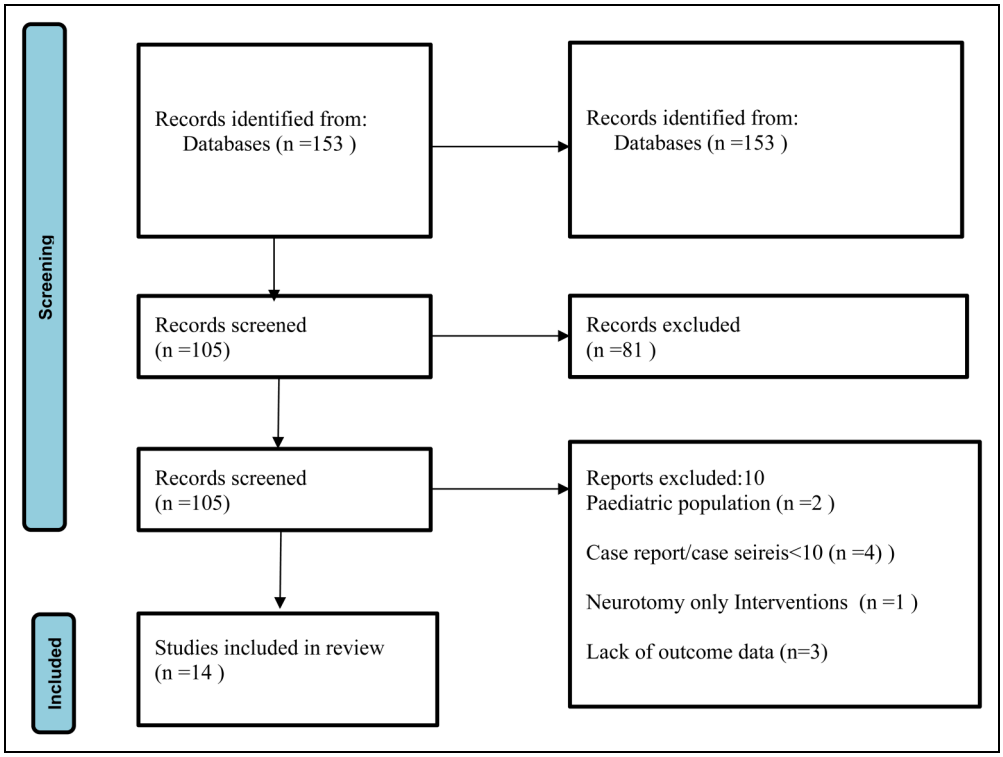

Figure 1. Depicts PRISMA flow diagram.

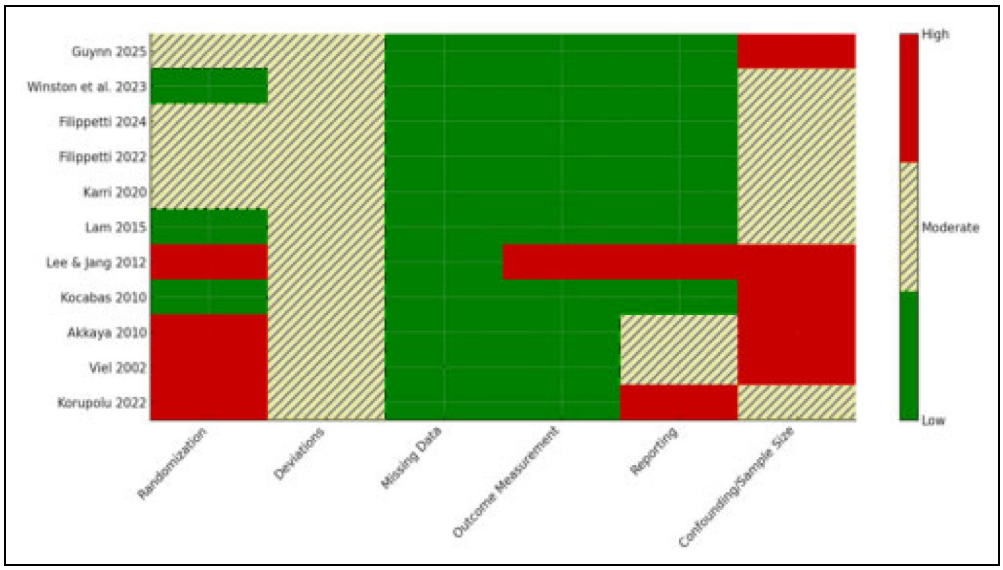

Figure 2. Depicts risk of bias of the therapeutic and diagnostic studies included in the systematic review.
Risk-of-bias assessment across included therapeutic, diagnostic, and cryoneurolysis studies. Domains are rated as low risk (horizontal banding), moderate risk (diagonal hatching), or high risk (solid dark fill). Most studies demonstrated low risk in outcome measurement and reporting, with higher variability in randomization, deviations from intended interventions, and sample size adequacy.

Risk of Bias Assessment

Overall methodological quality across the included studies was moderate. The 2 Randomized controlled trials (RCT) in the review demonstrated low risk of bias in sequence generation and outcome reporting but moderate risk related to allocation concealment and blinding, given the procedural nature of nerve block interventions (Kocabas et al., 2010; Lam et al., 2015). Most retrospective and case series studies (Filippetti et al., 2022, 2024a, 2024b; Lee & Jang, 2012) were rated as having moderate to high risk of bias, primarily

due to non-randomized design, small sample sizes, and potential selection or performance bias. However, outcome measurement bias was generally low. Nearly all the studies employed standardized clinical outcomes such as the Modified Ashworth Scale (MAS) and Goal Attainment Scale (GAS). No major issues with selective reporting or incomplete outcome data were identified. The assessments are depicted in figure form for all domains in Figure 2.

Data Synthesis

Due to heterogeneity of populations, interventions, and outcome measures, a narrative synthesis was performed. Results were summarized in following structured tables and figures:

- Study characteristics and interventions.
- Outcomes by indication and agent used.
- Safety profile.
- Trends over time (guidance methods, agents used).

Results

Study Selection

The database search yielded 153 records (PubMed 107, Embase 10, CINAHL 35, Cochrane 1). After the removal of 48 duplicates, 105 unique studies remained. Following title and abstract screening, 81 records were excluded. Full-text review was performed for 24 studies, of which 9 were excluded. A total of 14 studies spanning from 2002 to 2025 that investigated diagnostic, therapeutic, and cryoneurolytic nerve block techniques for spasticity management were selected. The study selection process is shown in the PRISMA flow diagram (Figure 1). Studies were included if they involved participants with spasticity due to upper motor neuron lesions (e.g., stroke, spinal cord injury, or mixed etiologies) and evaluated peripheral nerve block procedures using diagnostic local anaesthetic blocks, chemical neurolysis (phenol or alcohol), or cryoneurolysis. These studies collectively represent a broad international distribution across Europe, Asia, and North America and encompass both upper and lower limb spasticity presentations. Table 1 shows the characterstics of all the included studies for the review.

Trends Over Time

Interventions evolved from alcohol/phenol neurolysis to diagnostic/local anaesthetic blocks, with cryoneurolysis emerging in the recent years. Figures 3 illustrate temporal trends of utilization of nerve block modality in spasticity management. Landmark-guided injections dominated earlier studies, while more recent studies predominantly used ultrasound ± electrical stimulation guidance (Figure 4).

Diagnostic Nerve Blocks

The DNB studies primarily evaluated the predictive and functional value of temporary nerve blockade for spasticity management. A retrospective study reported that tibial diagnostic nerve block predicted a positive response to BoNT injections on improvement in ankle dorsiflexion following treatment in post-stroke equinovarus foot (Picelli et al., 2020). Another retrospective study of 46 patients found that diagnostic tibial blocks are a valuable tool for prediction of selective neurotomy outcomes in patients with spastic equinovarous foot (Lamora et al., 2023). Recent studies have further demonstrated that pre-BoNT tibial diagnostic block helped identify the specific muscle causing clonus and also can guide further regarding the BoNT doasage to these muscles which led to significantly reduced clonus frequency (–1.3 beats, $p < .05$) (Filippetti et al., 2024b), while in another study they confirmed higher Goal Attainment Scale (GAS) scores in patients receiving diagnostic blocks compared with controls (70 vs 40, $p = .028$) (Filippetti et al., 2024a). In another study they also observed an inverse correlation between muscle echo intensity and post-block outcome (Spearman's $\rho = -0.327$ to -0.352, $p = .008$–$.045$), indicating that patients with lower echo intensity showed greater tone reduction (Filippetti et al., 2022). Collectively, these studies show that DNBs yield quantifiable improvements in spasticity and functional outcomes, supporting their predictive value for individualized BoNT treatment (Table 2).

Therapeutic Nerve Blocks

Therapeutic neurolysis studies evaluated chemical agents such as phenol and alcohol for long-term tone reduction. Ethanol neurolysis of the obturator nerve was evaluated in 23 patients and showed significant improvements in adductor muscle spasticity ($p < .001$), gait ($p < .02$), hygiene ($p < .01$), and triple flexion ($p < .05$) (Viel et al., 2002). Another study showed that obturator nerve phenol neurolysis significantly improved hip abduction range, reduced Ashworth and VAS pain scores ($p < .005$), and enhanced hygiene scores for up to two months post-procedure (Akkaya et al., 2010). A RCT demonstrated decrease in MAS ($p < .001$) and reduced clonus beats ($p < .01$), and improved ankle dorsiflexion improved from $-18° \pm 6°$ to $-6° \pm 5°$ ($p < .05$) upto 6 months with tibial nerve phenol and alcohol neurolysis (Kocabas et al., 2010). Greater MAS reduction in phenol obturator neurolysis versus placebo ($p < .01$) and also decreased elbow flexor tone and improved extension after alcohol musculocutaneous neurolysis was reported in other studies (Lam et al., 2015; Lee & Jang, 2012). Further observational analyses confirmed immediate spasticity

Table 1. Characteristics of Included Studies (n = 14).

Author/Year	Design	Country	Diagnosis / Population	Sample Size (n)	Intervention (Nerve/ agent)	Key Outcomes
Viel et al., 2002	Case series	France	Post-stroke hip adductor spasticity	23	Ethanol obturator neurolysis	Reduced adductor tone; improved range of motion
Akkaya et al., 2010	Case series	Turkey	Severe hip adductor spasticity (SCI/ mixed)	62	Phenol obturator neurolysis	Significant MAS reduction; transient sensory symptoms <10%
Kocabas et al., 2010	RCT	Turkey	Post-stroke equinovarus foot	20	Phenol vs alcohol tibial neurolysis	Both agents ↓ MAS (−1.2 at 12 wks, $p < .001$)
Lee & Jang, 2012	Case series	South Korea	Hemiparetic stroke, elbow flexor spasticity	10	Alcohol musculocutaneous neurolysis	Reduced tone and improved elbow extension
Lam et al., 2015	RCT	Hong Kong	Hip adductor spasticity (LTC, stroke)	26	Phenol obturator vs placebo	Greater MAS reduction in phenol group ($p < .01$), improved GAS scores
Korupolu et al., 2022	Observational case series	USA	Spinal Cord Injury	66	Phenol- Obturator (33%), pectoral (23%), and other peripheral motor nerves	Spasticity improved in ~92% of encounters; immediate tone reduction reported in most cases
Karri et al., 2020	Retrospective cohort	India	Upper limb distal spasticity (stroke/ mixed)	57	Phenol neurolysis (median/ulnar/radial)	Safety outcomes-No persistent dysesthesia, neuropathic pain, or motor loss; only transient injection discomfort
Picelli et al., 2020	Retrospective cohort	Italy	Equinovarus foot (post-stroke)	52	Tibial diagnostic nerve block	Predicted BoNT response; improved dorsiflexion
Filippetti et al., 2022	Retrospective cohort	Italy	Post-stroke equinovarus foot	30	Tibial diagnostic block + echo intensity with LA 2 ml(MBB)	Inverse correlation ($\rho = -0.327$ to -0.352, $p = .008$–$.045$)
Lamora et al., 2023	Retrospective cohort	Belgium	Equinovarus foot (post-stroke)	46	Diagnostic tibial nerve block with LA 1 ml(MBB)	Supported decision for surgery
Filippetti et al., 2024a	Retrospective cohort	Italy	Equinovarus foot (post-stroke)	91	Tibial diagnostic block + BoNT with LA 2 ml(MBB)	Predicted clonus reduction (−1.3 beats, $p < .05$)
Filippetti et al., 2024b	Case–control	Italy	Equinovarus foot (post-stroke)	42	Diagnostic tibial block prior to BoNT-A with LA 2 ml	Higher GAS scores (58 ± 9 vs. 49 ± 11, $p = .01$)
Winston et al., 2023	Prospective multicentre cohort	USA	Mixed limb spasticity (various neurological etiologies)	113	Percutaneous cryoneurolysis (multiple peripheral motor nerves; 277 nerves treated) with LA 2 ml (MBB)	Safety outcomes: 96.75% of nerve treatments had no post-procedure pain or dysesthesia; adverse events were rare and mild (transient soreness, bruising); no serious neurological complications reported
Guynn, 2025	Case series	USA	Spasticity (mixed etiologies)	10	Percutaneous cryoneurolysis	Reduced MAS; no major adverse events

MBB = motor branch block.

reduction in approximately 92% of procedures among individuals with spinal cord injury with ultrasound- and stimulation-guided phenol neurolysis with no adverse events was also observed (Korupolu et al., 2022) (Table 3).

These various study findings support neurolysis as effective alternative therapeutic strategy where other treatment modalities might not be the ideal option for focal spasticity management.

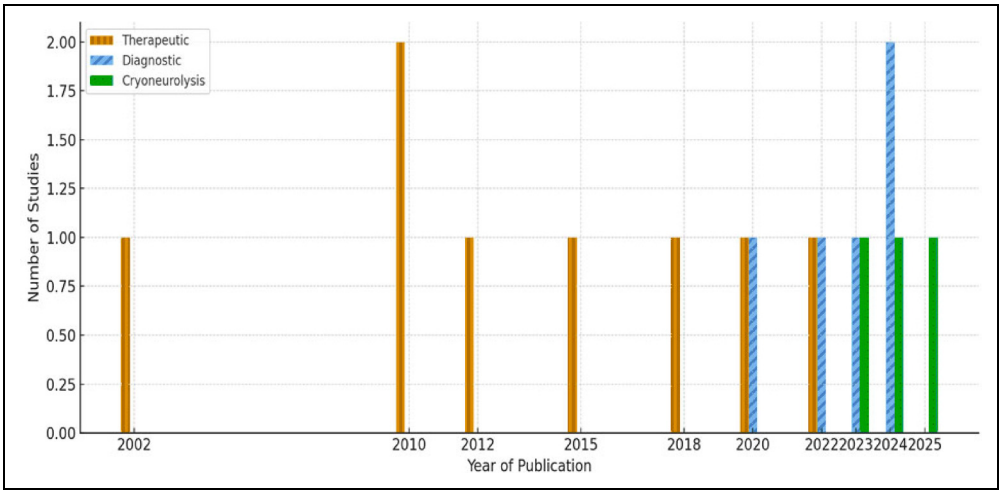

Figure 3. Summary of trend over time for nerve block studies in management of spasticity (2002–2025).
Early publications were dominated by therapeutic neurolysis, whereas studies after 2018 increasingly examined diagnostic nerve blocks and cryoneurolysis, reflecting diversification of interventional approaches for spasticity.

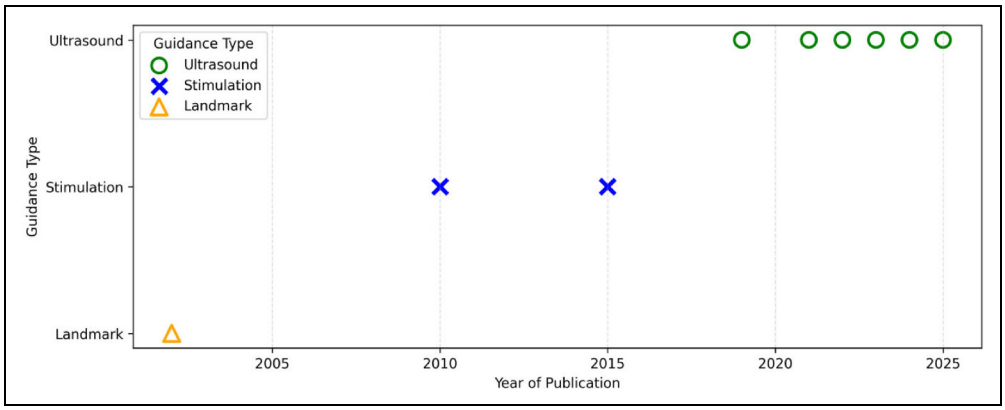

Figure 4. Summary of use of guidance technique for nerve blocks over the years (2002–2025).
Guidance methods used across included nerve block studies. Landmark-based techniques were used in earlier studies, stimulation guidance peaked around 2010–2015, and ultrasound guidance became the predominant modality in more recent publications.

Cryoneurolysis

A case series documented consistent MAS reduction in 10 patients treated with percutaneous cryoneurolysis, without major adverse events (Guynn, 2025) (Table 4). This suggests that cryoneurolysis provides measurable, durable tone reduction comparable to chemical neurolysis, with favorable safety and functional profile.Another prospective study based on safety outcomes of this newer modality showed 96.75% of nerve treatments had no post-procedure pain or dysesthesia (Wincent et al., 2023).

Effective Dosage

Across therapeutic neurolysis studies, effective doses were remarkably consistent, with obturator neurolysis typically using 5–10 mL (Akkaya et al., 2010; Viel et al., 2002), tibial neurolysis using approximately 5 mL (Kocabas et al., 2010), and upper-limb phenol neurolysis requiring smaller targeted volumes of 1–3 mL per nerve (Karri et al., 2020), collectively indicating that clinically effective tone reduction is achieved within a predictable low-volume range tailored to nerve size. Diagnostic blocks were generally done to motor branches with 1–2 ml with local anaesthetic.

Duration of Effect

Across the therapeutic studies included in this review, neurolytic blocks consistently demonstrated longer and more predictable durations of effect, with obturator and musculocutaneous phenol neurolysis producing clinically

Table 2. Depicts Results from Diagnostic Nerve Block (DNB) Studies.

S.no	Study (Year)	Primary Outcome Measures	Key Findings	Agent/Dose	Adverse Event
1.	Picelli et al., 2020	MAS, Passive ankle range Tardieu scale	Significant reduction in MAS (−1.2 ± 0.4, $p < .001$) after DNB; Tardieu spasticity grade −1.0, improved passive ROM and gait symmetry	with LA (1–2 ml for MBB 5–6 ml for main trunk)	None
2.	Filippetti et al., 2022	MAS, Echo intensity	MAS decreased (−1.0 ± 0.3, $p < .01$); reduction correlated with lower muscle echo intensity (r = −0.64)	with LA (1–2 ml for MBB)	None
3.	Lamora et al., 2023	MAS, ROM	MAS significantly reduced after DNB and selective neurotomy ($p < .001$); increased active dorsiflexion (XVA + 10°)	with LA (1 ml for MBB)	None
4.	Filippetti et al., 2024a, 2024b	Tardieu clonus grade	Median clonus grade improved from 3 → 2 after DNB ($p < .05$); guided targeted BoNT-A injection	with LA (1–2 ml for MBB)	None
5.	Filippetti et al., 2024a, 2024b	Goal Attainment Scale (GAS-light), MAS	70% goal achievement vs. 40% in controls ($p < .05$); improved functional outcomes	with LA (1–2 ml for MBB)	None

LA = local anaesthetic; MBB = motor branch block; XVA = active range of motion.

meaningful tone reduction for 3–6 months (Akkaya et al., 2010; Lam et al., 2015; Viel et al., 2002). In contrast, the duration of benefit from botulinum toxin A is both shorter and more variable. Pharmacodynamic analyses show peak effect at 4–6 weeks followed by progressive decline, with most patients returning toward baseline by 10–12 weeks (Gill et al., 2024; Doan et al., 2021). A large meta-analysis also reports that BoNT-A often provides only 2–3 months of meaningful spasticity reduction and that the duration varies substantially between muscles and patients (Ojardias et al., 2022). This comparative pattern highlights that neurolysis offers a more sustained modulation of spasticity, whereas BoNT-A requires more frequent repeat cycles and exhibits greater inter-individual variability in its clinical effect.

Functional Outcomes Across Nerve Block Interventions

Across the included studies, functional outcomes were rarely reported. Obturator neurolysis trials (Akkaya et al., 2010; Viel et al., 2002) focused primarily on impairment-level changes, such as reductions in adductor tone and improved passive range of motion (ROM), with functional implications like reduced scissoring in gait or improved hygiene access discussed only conceptually rather than qualitatively measured. Similarly, a randomized trial demonstrated significant MAS reductions following phenol obturator neurolysis, but did not include gait, ADL, or GAS-based functional assessments. However it can be associated with improved mobility and easier hygiene care (Lam et al., 2015). In contrast, most tibial neurolysis and diagnostic block studies (Filippetti et al., 2022; Kocabas et al., 2010; Lamora et al., 2023; Picelli et al., 2020) focused mainly on impairment-level outcomes, with functional relevance inferred from biomechanical changes such as improved ankle dorsiflexion, reduced clonus, or better muscle targeting to support gait . Only one recent case control study reported quantifiable activity-level outcomes, showing that incorporating a diagnostic tibial nerve block improved GAS achievement for patient-centred goals such as safer walking and easier transfers—with approximately 70% of goals met compared to 40% in patients assessed clinically alone. (Filippetti et al., 2024b). Upper limb neurolysis studies (Karri et al., 2020; Lee & Jang, 2012) demonstrated reductions in spasticity and increased ROM but did not quantify functional upper-limb goals. In the recent cryoneurolysis study, clinically meaningful functional improvements were reported, including better gait alignment (reduced internal foot rotation, improved heel contact), smoother mobility with less spastic interference, enhanced limb use, and improved patient-reported daily functioning, despite the absence of standardized functional scales (Guynn, 2025).

Safety

Safety outcomes were generally well described across studies, with no reports of serious or permanent adverse events. Among therapeutic neurolysis studies, transient sensory symptoms were reported in fewer than 10% of participants and mild local pain or dysaesthesia was occasionally noted after phenol or alcohol injection (Akkaya et al., 2010; Kocabas et al., 2010 ; Lam et al., 2015). No major vascular, motor, or systemic complications were observed. DNB studies uniformly reported no adverse events, confirming the safety of low-dose local anaesthetic use for prognostic assessment (Filippetti et al., 2022, 2024a, 2024b; Picelli et al., 2020). In cryoneurolysis studies mild

Table 3. Depicts Results from Therapeutic Nerve Block (TNB) Studies.

S. No.	Study (Year)	Target Nerve	Agent & Volume	Outcome Measures	Results	*p*-Values	Adverse Events	Duration of Effect
1.	Lee & Jang, 2012	Musculocutaneous nerve	Alcohol 35% (2.5–4 ml)	MAS, elbow ROM, FIM	MAS ↓ 1–2 grades; ROM ↑ 15°	$p < .01$	None	> 3 months
2.	Karri et al., 2020	Median / Ulnar / Radial	Phenol 5%, 1.8 mL/nerve (range 0.5–9.0 mL)	Safety outcomes	—	—	No persistent dysesthesia, neuropathic pain, or motor loss; only transient injection discomfort	2–4 months
3.	Korupolu et al., 2022	Obturator, pectoral, and mixed UL/LL nerves	Phenol 6%-1.5 mL/nerve (median), 4 mL total	MAS; clinician-reported tone	Improvement in ~92% encounters	Not reported	None	12 weeks
4.	Viel et al., 2002	Obturator nerve	Ethanol 45–50%, 8–9 mL (4 mL anterior; 4–5 mL posterior)	MAS, gait, hygiene, triple-flexion	Adductor spasticity significantly reduced ($p < .001$ at 1 day; $p < .01$ at 60–120 days); triple-flexion response improved ($p < .05$ from days 1–120); gait improved ($p < .02$) and hygiene scores enhanced ($p < .01$).	MAS-$p < .001$; gait $p < .02$	None	Up to 12 weeks
5.	Akkaya et al., 2010	Obturator nerve	Phenol 6%, 5–10 mL	MAS, ROM, hygiene, VAS	MAS ↓ 1.4 ± 0.3; ↑ abduction; ↓ VAS; hygiene improved	$p < .005$	Mild transient pain (6%)	8–12 weeks
6.	Lam et al., 2015	Obturator nerve	Phenol 5%	MAS, GAS scores, hygiene	MAS ↓ 1.2; hygiene improved	$p < .01$	Mild pain	≈ 6 months
7.	Kocabas et al., 2010	Tibial nerve	Phenol 5% or Alcohol 50%, 5 mL	MAS, clonus, ROM, gait	MAS 3.2→1.8; clonus 4.3→1.6; ROM −18°→−6°	MAS $p < .001$; clonus $p < .01$	Mild dysesthesia (10%)in alcohol group	Weeks to months (variable)

MAS = modified ashworth scale; ROM = range of motion; VAS = visual analog scale; GAS = goal attainment scaling; UL = upper limb; LL = lower limb; SCI = spinal cord injury; RCT = randomized controlled trial; TNB = therapeutic neurolytic block; BoNT-A = botulinum neurotoxin type A; MCN = musculocutaneous nerve; mL = millilitre.

Table 4. Shows Results for Cryoneurolysis Studies.

S.no	Study (Year)	Intervention	Outcome Measures	Results	p-Values	Adverse Events
1	Winston et al., 2023	Percutaneous cryoneurolysis (277 nerve treatments across upper & lower limbs)	Safety profile; incidence of post-procedure pain, dysesthesia, complications	96.75% of nerve procedures had no pain or dysesthesia beyond treatment period; overall complication rate very low	Not applicable (descriptive safety study)	Mild transient soreness or bruising in a few cases; no serious neurological adverse events
2..	Guynn, 2025	Cryoneurolysis (upper limb)	MAS, pain (VAS)	MAS ↓ 1.0; VAS ↓ 30%; no motor loss	$p < .05$	None

bruising and transient soreness were described in a small proportion of cases, but no infections or long-term sensory loss were reported (Guynn, 2025; Winston et al., 2023). This suggests that all nerve block modalities exhibit a favorable safety profile, with minor, self-limiting, and complications (Figure 5). Table 5 summarizes the comparitive features of different nerve blocks in the included studies.

Discussion

This systematic review summarizes the available evidence on diagnostic and therapeutic nerve blocks for spasticity published since 2000, including phenol and alcohol neurolysis, cryoneurolysis, and DNBs used alone or in combination with BoNT. The overall findings support nerve blocks as a valuable intervention, both for therapeutic use for spasticity reduction and for guiding individualized treatment strategies.

Early studies primarily focused on chemical neurolysis using phenol or alcohol to achieve long-term denervation in focal spasticity. These interventions produced significant reductions in MAS scores, often sustained for several months, with only minor transient sensory effects (Kocabas et al., 2010; Lee & Jang, 2012; Viel et al., 2002). Although most of these studies were case series or small randomized trials, they established phenol neurolysis as a durable treatment option, particularly in settings where botulinum toxin (BoNT) therapy is limited or insufficient.

From 2018 onward, more research has been directed towards DNB aimed at predicting response to BoNT injections or guiding surgical decisions. Various studies highlighted the clinical utility of local anaesthetic blocks for individualized treatment planning (Filippetti et al., 2022, 2024a; Lamora et al., 2023; Picelli et al., 2020). Objective correlations between diagnostic block outcomes and muscle ultrasound characteristics (echo intensity), as well as improvements in clonus reduction and Goal Attainment Scale (GAS) scores, underscore the predictive and goal-oriented value of this approach (Filippetti et al., 2022, 2024a, 2024b). These blocks help to establish if spasticity is the primary reason of functional or activity limitation and help localise the key muscle involved. Improvements following DNB support escalation to botulinum toxin or therapeutic neurolysis. This can help with more efficient use of BoNT. No improvement in tone or function with these blocks suggest fixed contracture or structural pathology and may indicate the need for surgical intervention or alternative rehabilitation strategies.

More recent studies have explored cryoneurolysis as a minimally invasive and a safe alternative for tone reduction (Guynn, 2025; Winston et al., 2023). While cryoneurolysis is can be considered as a promising emerging modality for the management of spasticity, the current evidence remains limited. Most published data are derived from case series, feasibility studies, and early safety evaluations, with a lack of robust comparative or long-term outcome studies.

From an ICF-based activity and participation perspective, the available evidence suggests that functional gains following nerve blocks for spasticity are selective rather than universal. Improvements at the activity level were most consistently reported following obturator neurolysis and when diagnostic tibial nerve blocks were used as part of structured, goal-based treatment planning. These findings indicate that nerve blocks may be most effective in enhancing activity performance when carefully targeted to address spasticity patterns that directly constrain task execution, such as standing, walking, or personal care activities. Importantly, nerve blocks should not be viewed solely as interventions targeting tone reduction. By selectively weakening overactive muscle groups, nerve blocks can improve limb alignment, reduce pain and involuntary muscle activity, and optimise biomechanical conditions necessary for efficient movement. These changes at the impairment level can translate into meaningful activity-level benefits, such as improved gait mechanics, easier transfers, improved limb positioning for hygiene or dressing, and increased tolerance of orthotic devices. Furthermore, reduction in spasticity may create a therapeutic window in which individuals are better able to engage in rehabilitation interventions. Improved limb positioning and reduced resistance to

Table 5. Comparative Features of Nerve Block Techniques Based on Included Studies.

Block Type (From Included Studies)	Evidence of Effectiveness (From Review Studies)	Positive Clinical Indicators	Limitations
Therapeutic obturator neurolysis – ethanol / phenol (Akkaya et al., 2010; Lam et al., 2015; Viel et al., 2002)	Consistent MAS reduction of ~1–1.5 grades; improved hip abduction, gait base, hygiene and perineal care; effects maintained for ≥3–6 months in most cohorts.	Severe hip adductor spasticity with scissoring gait, difficulty with perineal hygiene, dressing and sitting positioning; patients in LTC or chronic stroke where BoNT alone is insufficient; goal is durable tone reduction rather than fine, selective control.	Evidence mainly from single-centre case series and small RCTs; mostly post-stroke, hip adductor–dominant patterns; phenol associated with mild transient pain in a minority (Akkaya et al.); limited data on long-term (>6–12 months) functional trajectories.
Therapeutic tibial neurolysis – phenol / alcohol (Kocabas et al., 2010)	RCT showed MAS reduction (3.2→1.8), reduced clonus and improved ankle ROM over ~12 weeks; both phenol and alcohol effective at impairment level.	Post-stroke equinovarus foot with clear dynamic plantarflexor overactivity; when reduction in equinus and clonus is needed to improve dorsiflexion, brace fitting or gait training.	Dysaesthesia occurred only in the alcohol group (~10%), not in the phenol group; small sample size pilot RCT; functional outcomes described qualitatively
Upper-limb neurolysis – musculocutaneous nerve (alcohol / phenol) (Lee & Jang, 2012; Matsumoto et al., 2018)	Both case series and retrospective cohort show significant elbow flexor MAS reduction with improved passive and, in some patients, active extension for ~8–12 weeks.	Hemiparetic elbow flexor spasticity limiting limb positioning, hygiene, dressing or arm swing, particularly when biceps overactivity dominates and other treatments plateau.	Very small numbers (Lee & Jang: n = 10); mostly impairment-level outcomes, limited formal activity data; duration modest (~2–3 months); alcohol carries theoretical sensory risk, though no major adverse events reported in these cohorts.
Upper-limb distal phenol neurolysis – median/ulnar/radial (Karri et al., 2020; Korupolu et al., 2022)	Safety outcomes in Karri et al.); Korupolu et al. report immediate spasticity improvement in ~92% of encounters across upper and lower limb nerves.	Focal distal upper-limb spasticity post-stroke or mixed aetiologies, where finger/wrist flexors or intrinsic hand spasticity interfere with grasp–release, hygiene, splinting or care; useful when specific nerve distributions can be clearly mapped.	Retrospective designs; heterogeneous populations (stroke + SCI + mixed); long-term functional and participation outcomes not well described; phenol volumes low but technique requires operator experience.
Diagnostic tibial nerve blocks (Filippetti et al., 2022, 2024a, 2024b; Lamora et al., 2023; Picelli et al., 2020)	Consistently produced transient reduction in spastic equinovarus and improved dorsiflexion; predicted BoNT response (Picelli et al.); associated with better goal attainment and higher GAS scores when used to plan BoNT (Filippetti et al., 2024b ~70% vs 40% goal achievement); helped stratify patients for surgery vs BoNT vs neurotomy (Lamora et al.).	Post-stroke equinovarus where it is unclear whether deformity is mainly dynamic or fixed; In patients to decide between BoNT, surgery or selective neurotomy; to test whether tone reduction actually improves gait, orthotic tolerance, or clonus before committing to an irreversible procedure.	Effects purely diagnostic and short-lived (minutes–hours); no therapeutic duration; all data from post-stroke equinovarus – generalisability to other patterns is uncertain; mostly retrospective, single-centre cohorts, with limited long-term functional follow-up.
Cryoneurolysis (Guynn, 2025; Winston et al., 2023)	Winston et al. (2023) multicentre cohort (n = 53) showed significant MAS reductions over several weeks with a low adverse-event rate; Guynn (2025) small case series (n = 10) reported MAS ↓ ~1 grade and ~30% pain reduction, with preserved motor function and ~12-week effect.	Mixed limb spasticity (including plateaued / refractory cases) where a reversible, medium-term nerve lesion is desired; useful when a prior diagnostic block suggests benefit, but there is a wish to avoid permanent chemical neurolysis; particularly attractive when pain or interference with function is a key goal and preservation of connective architecture is appealing.	Evidence still limited to cohort and case-series data; no RCTs against phenol or BoNT; follow-up generally ≤3 months; mainly upper-limb and mixed patterns; access to cryoneurolysis equipment and expertise may limit availability.

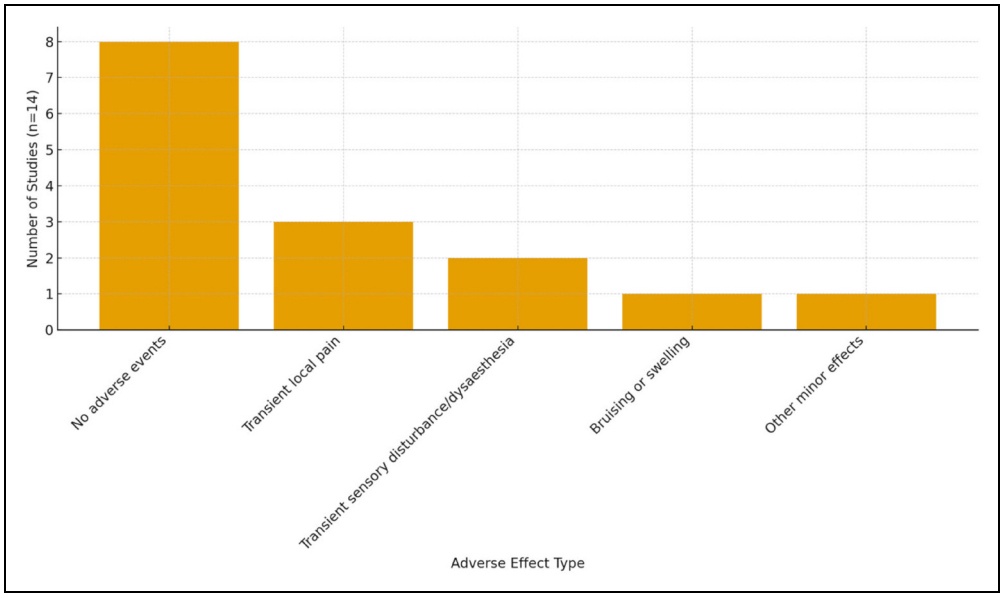

Figure 5. Depicts adverse effects across included studies.
Most studies reported no adverse events, while others noted only mild, transient issues such as local pain or temporary dysaesthesia. No serious or permanent complications were described.

movement can facilitate more effective stretching, strengthening, gait training, and motor relearning interventions that are often constrained when spasticity dominates posture or movement patterns. In this context, nerve blocks act as an enabling intervention, supporting engagement in task-specific practice and functional training.

At the participation level, these activity gains have the potential to support broader involvement in daily life roles, including increased independence in self-care, improved mobility within the home and community, and greater participation in rehabilitation programmes. However, the current evidence linking nerve blocks directly to participation outcomes remains limited, highlighting the importance of embedding nerve blocks within multidisciplinary, goal-oriented rehabilitation pathways to maximise their functional impact (Akkaya et al., 2010; Filippetti et al., 2024b; Royal College of Physicians, 2018).

The favorable safety record also strengthens confidence in the procedural role of nerve blocks within multidisciplinary spasticity management frameworks (Akkaya et al., 2010; Kocabas et al., 2010; Lam et al., 2015).

Another advantage of nerve blocks,either with of alcohol, phenol or local anaesthetics, is their cost effectiveness. Phenol and alcohol are inexpensive, readily available agents capable of producing comparable and often longer-lasting reductions in spasticity (Akkaya et al., 2010; Lam et al., 2015; Viel et al., 2002). In contrast, BoNT requires repeated administrations every three to four months, has dose limitations, and incurs higher cost per cycle (Dressler & Bhidayasiri, 2015).

Strengths and Limitations

Our review has several methodological strengths, we followed a preregistered protocol, applied comprehensive eligibility criteria, and conducted a systematic search across multiple databases. By restricting inclusion to studies published after 2000, we ensured that the findings reflect current practice with modern image-guidance techniques. The inclusion of both diagnostic and therapeutic interventions allowed for a broad yet clinically focused synthesis, while quantitative data extraction, including effect sizes and *p*-values, enhanced interpretability and clinical applicability. Study selection was transparent, with reasons for exclusion documented in accordance with PRISMA guidelines. However, several limitations should be acknowledged. Only fourteen studies met inclusion criteria, and most were small, single-centre cohorts with limited statistical power, making their findings vulnerable to random variation and restricting generalisability. Substantial heterogeneity was present across studies, including follow-up duration ranging from 12 weeks and up to 3–6 months in therapeutic neurolysis, involvement of multiple target nerves across upper and lower limbs, wide variation in agents and dosing protocols, and use of diverse outcome measures (primarily MAS, with inconsistent functional and goal-based assessments), precluding meaningful quantitative pooling and necessitating narrative synthesis. In addition, shorter follow-up periods also provided little insight into long-term durability or the need for repeat interventions. These limitations constrain the evidentiary

strength of the individual studies and highlight the value of a structured, systematic approach such as the present review, which applies consistent methodological criteria to synthesise an otherwise heterogeneous evidence. Finally, publication bias cannot be excluded, as grey literature and non-English studies were not systematically searched.

Conclusion

The overall findings indicate that nerve blocks are valuable interventions to be considered both for spasticity reduction and for guiding individualized treatment strategies.They can provide rapid and, in some cases, long-lasting tone reduction, this can then facilitate patients' rehabilitation leading to better functional outcomes. Diagnostic blocks improve the target muscle selection and serve as valuable prognostic tools by identifying patients most likely to respond to botulinum toxin or neurolytic procedures, this in turn can leads to more cost-effectiveness. The emerging modalities such as cryoneurolysis have also shown to offer minimally invasive and reversible interventions with promising early results. These features underscore the importance of continued research into nerve blocks as a componenet of multimodal spasticity care. By modulating impairments with good control of spasticity, nerve blocks may therefore serve as valuable adjuncts that facilitate functional gains and support long-term, goal-directed rehabilitation planning.

Future research with well-designed multicenter randomized controlled trials should aim at comparing phenol, alcohol, and cryoneurolysis with BoNT, using standardized outcomes such as the MAS, Tardieu Scale, Goal Attainment Scaling, and quality of life measures. Long-term follow-up studies can be done to evaluate the durability of effects and outcomes of repeat procedures. This review highlights the importance of Nerve Blocks in multimodal spasticity care, particularly in resource-limited settings.

ORCID iD

Sagarika Muradia https://orcid.org/0009-0008-1128-4893

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Supplemental Material

Supplemental material for this article is available online.

References

Akkaya, T., Unlu, E., Alptekin, A., Gumus, H. I., Umay, E., & Cakci, A. (2010). Neurolytic phenol blockade of the obturator nerve for severe adductor spasticity. *Acta Anaesthesiologica Scandinavica*, *54*(1), 79–85. https://doi.org/10.1111/j.1399-6576.2009.02130.x

Banawal, L. K., Chandrol, V., Ratre, S., Yadav, Y. R., & Bajaj, J. (2025). Surgical approach for spasticity: A systematic review. *Indian Journal of Neurosurgery*, *14*(Suppl S1), S7–S19. https://doi.org/10.1055/s-0045-1810050

Corazza, A. (2020). *Shear wave elastography to assess the effect of botulinum toxin in muscle hypertonia following stroke* [PhD thesis]. University of Padova.

Doan, Q. V., Brashear, A., Elovic, E., McIntyre, A., & O'Dell, M. W. (2021). Efficacy and optimal dose of botulinum toxin A in post-stroke lower-extremity spasticity: A systematic review and meta-analysis. *Toxins*, *13*(6), 428. https://doi.org/10.3390/toxins13060428

Dressler, D., & Bhidayasiri, R. (2015). Botulinum toxin therapy of spasticity: Quality of life and treatment expectations. *European Journal of Neurology*, *22*(6), 815–821. https://doi.org/10.1111/ene.12666

Filippetti, M., Di Censo, R., Varalta, V., Baricich, A., Santamato, A., Smania, N., & Picelli, A. (2022). Is the outcome of diagnostic nerve block related to spastic muscle echo intensity? *Journal of Rehabilitation Medicine*, *54*, jrm00275. https://doi.org/10.2340/jrm.v54.85

Filippetti, M., Tamburin, S., Di Censo, R., Adamo, M., Manera, E., Ingrá, J., Mantovani, E., Facciorusso, S., Battaglia, M., Baricich, A., Santamato, A., Smania, N., & Picelli, A. (2024a). Role of diagnostic nerve blocks in goal-oriented treatment of spasticity with botulinum toxin type A: A case–control study. *Toxins*, *16*(6), 258. https://doi.org/10.3390/toxins16060258

Filippetti, M., Tamburin, S., Di Maria, I., Angeli, C., Di Censo, R., Mantovani, E., Smania, N., & Picelli, A. (2024b). Diagnostic nerve block to guide botulinum neurotoxin type A injection for clonus in spastic equinovarus foot: A retrospective study. *Toxins*, *16*(12), 503. https://doi.org/10.3390/toxins16120503

Gill, R., Banky, M., Yang, Z., Medina Mena, P., Woo, C. C. A., Bryant, A., Olver, J., Moore, E., & Williams, G. (2024). The Effect of Botulinum Neurotoxin-A (BoNT-A) on Muscle Strength in Adult-Onset Neurological Conditions with Focal Muscle Spasticity: A Systematic Review.. *Toxins*, *16*(8), 347. https://doi.org/10.3390/toxins16080347

Guynn, C. C. (2025). Percutaneous cryoneurolysis as a dynamic treatment for spasticity: A case series. *Archives of Rehabilitation Research and Clinical Translation*, *7*(1), 100426. https://doi.org/10.1016/j.arrct.2025.100426

Heinen, F., Desloovere, K., Schroeder, A. S., Berweck, S., Borggraefe, I., van Campenhout, A., Andersen, G. L., Aydin, R., Becher, J. G., Bernert, G., Caballero, I. M., Carr, L., Valayer, E. C., Desiato, M. T., Fairhurst, C., Filipetti, P., Hassink, R. I., Hustedt, U,, Jozwiak, M., … Molenaers, G. (2010). The updated European Consensus 2009 on theuse of

Botulinum toxin for children with cerebral palsy. *European Journal of Paediatric Neurology*, 14(1), 45–66. https://doi.org/10.1016/j.ejpn.2009.09.005

Karri, J., Zhang, B., & Li,S. (2020). Phenol neurolysis for management of focal spasticity in the distal upper extremity. *PM & R: The Journal of Injury, Function, and Rehabilitation*, *12*(3), 246–250. https://doi.org/10.1002/pmrj.12217

Kocabas, H., Salli, A., Demir, A. H., Ozerbil, O. M., & Ugurlu, H. (2010). Comparison of phenol and alcohol neurolysis of tibial nerve motor branches to the gastrocnemius muscle for treatment of spastic foot after stroke: A randomized controlled pilot study. *European Journal of Physical and Rehabilitation Medicine*, *46*(1), 5–10. PMID: 20332720.

Korupolu, R., Malik, A., Pemberton, E., Stampas, A., & Li, S. (2022). Phenol neurolysis in people with spinal cord injury: A descriptive study. *Spinal Cord Series and Cases*, *8*, 90. https://doi.org/10.1038/s41394-022-00556-0

Lam, K., Wong, D., Tam, C. K., Wah, S. H., Myint, M. W., Yu, T. K., So, K. K., Cheung, G., Au, K. M., Fu, M.H., Wu, Y. M., & Kng, C. P. (2015). Ultrasound and electrical stimulator-guided obturator nerve block with phenol in the treatment of hip adductor spasticity in long-term care patients: A randomized, triple blind, placebo controlled study. *Journal of the American Medical Directors Association*, *16*(3), 238–246. https://doi.org/10.1016/j.jamda.2014.10.005

Lamora, J.-P., Deltombe, T., & Gustin, T. (2023). Effects of diagnostic tibial nerve block and selective tibial neurotomy on spasticity and co-contractions: A retrospective observational study. *Journal of Rehabilitation Medicine*, *55*, jrm4850. https://doi.org/10.2340/jrm.v55.4850

Lee, D. G., & Jang, S. H. (2012). Ultrasound-guided alcohol neurolysis of the musculocutaneous nerve for elbow spasticity in hemiparetic stroke. *NeuroRehabilitation*, *31*, 373–377. https://doi.org/10.3233/NRE-2012-00806

Matsumoto, M. E., Berry, J., Yung, H., Matsumoto, M., & Munin, M. C. (2018). Comparing electrical stimulation with and without ultrasound guidance for phenol neurolysis to the musculocutaneous nerve. *PM & R : The Journal of Injury, Function, and Rehabilitation*, *10*(4), 357–364. https://doi.org/10.1016/j.pmrj.2017.09.006

National Institute for Health and Care Excellence (NICE). (2023). Economic model: cost-utility analysis of botulinum toxin A to reduce spasticity. In *Stroke rehabilitation in adults (NG236)* (pp. 51–54). NICE.

Ojardias, E., Ollier, E., Lafaie, L., Celarier, T., Giraux, P., & Bertoletti, L. (2022). Time course response after single injection of botulinum toxin to treat spasticity after stroke: Systematic review with pharmacodynamic model-based meta-analysis. *Annals of Physical and Rehabilitation Medicine*, *65*(3), 101579. https://doi.org/10.1016/j.rehab.2021.101579

Picelli, A., Battistuzzi, E., Filippetti, M., Modenese, A., Gandolfi, M., Munari, D., & Smania, N. (2020). Diagnostic nerve block in prediction of outcome of botulinum toxin treatment for spastic equinovarus foot after stroke. *Journal of Rehabilitation Medicine*, *52*, jrm00069. https://doi.org/10.2340/16501977-2693

Royal College of Physicians. (2018). *Spasticity in adults: management using botulinum toxin* (2nd ed.). RCP.

Trompetto, C., Marinelli, L., Mori, L., Pelosin, E., Currà, A., Molfetta, L., & Abbruzzese, G. (2014). Pathophysiology of spasticity: Implications for neurorehabilitation. *BioMed Research International*, *2014*, 354906. https://doi.org/10.1155/2014/354906

Viel, E. J., Perennou, D., Ripart, J., Pélissier, J., & Eledjam, J. J. (2002). Neurolytic blockade of the obturator nerve for intractable adductor spasticity. *European Journal of Pain*, *6*(2), 97–104. https://doi.org/10.1053/eujp.2001.0269

Winston, P., MacRae, F., Rajapakshe, S., Morrissey, I., Boissonnault, É., Vincent, D., & Hashemi, M. (2023). Analysis of adverse effects of cryoneurolysis for the treatment of spasticity. *American Journal of Physical Medicine & Rehabilitation*, *102*(11), 1008–1013. https://doi.org/10.1097/PHM.0000000000002267

Copyright of NeuroRehabilitation is the property of Sage Publications Inc. and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.